Biochemistry Cumulative Examination

SEPTEMBER 5, 2003

1. The twenty major amino acids normally found in proteins are listed below. Each is different from the others, but still they can be grouped by their properties. In your blue book, next to each amino acid name write the appropriate letter code(s) which describe(s) the properties of that amino acid. 58 points, 2 for each correct answer (a hint as to the number of letters you should write down).

Alanine	Leucine
Arginine	Lysine
Asparagine	Methionine
Aspartate	Phenylalanine
Cysteine	Proline
Glutamate	Serine
Glutamine	Threonine
Glycine	Tryptophan
Histidine	Tyrosine
Isoleucine	Valine

A. Hydrophobic aliphatic
B. Hydrophobic aromatic
C. Hydrophilic neutral
D. Hydrophilic positively charged
E. Hydrophilic negatively charged
F. Sulfur containing
G. Achiral
H. Multiple chiral centers
I. Secondary amine

2. What is the difference between cysteine and cystine (besides the letter e)? 10 points

3. Name each amino acid that has a side chain that can be ionized in the pH range of 2-11. Next to each name give the pK_a (within ±1 pH unit). 15 points

4. Amino acid analysis can be carried out to determine the relative ratio of the different amino acids in proteins. However, several amino acids can not be distinguished from each other in a protein hydrolysate. Explain which amino acids these are and why this is so. 15 points

5. Name the smallest amino acid and the largest. 2 points

Biochemical Cumulative Examination
December 5, 2003
Enzymes

1. For the following enzyme reaction

$$E+S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E+P$$

a. Derive the Michaelis-Menten equation (25%)

$$v = \frac{V_{\max}[S]}{K_m + [S]}, \text{ where } K_m = \frac{k_{-1}+k_2}{k_1}$$

b. Explain the significance of the following terms (25%)

(i) $k_{cat}(=k_2)$

(ii) K_m

(iii) k_{cat}/K_m

(iv) $V_{\max}$

(v) turnover number

2. In terms of the change of quaternary structure in hemoglobin, explain the cause of cooperativity of hemoglobin in O_2 binding. (25%)

3. Describe the allostery effect when Trp receptor interacts with Trp operon DNA. (25%)